Unit 2 - Overview of Excel for Data Visualization

- Introduction
  - Overview of Data Visualization in Excel
    - Excel is a widely used tool for organizing, analyzing, and visualizing data across various industries.
    - The unit introduces the fundamentals of Excel, focusing on its interface and workspace.
    - Key components like the Ribbon, menus, and toolbars enable efficient navigation and utilization of Excel.
- Introduction to Excel
  - Overview of Microsoft Excel
    - Excel is a powerful and versatile spreadsheet application introduced in 1985.
    - Supports various platforms including Windows, macOS, Android, and iOS.
    - Known for its extensive features including data visualization, statistical analysis, and basic machine learning.
  - Excel Interface Essentials
    - Standard workbook consists of worksheets containing rows and columns, forming cells for data entry.
    - Offers depth of functionality for both beginner and advanced users, such as pivot tables and VBA scripting.
    - Importance in the modern world across sectors such as business, healthcare, academia, and personal use.
  - Features of Excel
    - Data Storage and Management
    - Mathematical Functions
    - Data Visualization Tools
    - Pivot Tables
    - Macro and Scripting
    - Collaboration Features
  - Limitations of Excel
    - Row and column limits per worksheet.
    - Lacks relational capabilities of databases.
    - Steep learning curve for advanced features.
  - Versions and Pricing
    - Different versions offer incremental improvements and varying pricing models.
- Spreadsheet Basics
  - Definition and Core Elements
    - Spreadsheet as a digital canvas with rows, columns, and cells for data organization.
    - Key elements: Cell, Row, Column, Worksheet, and Workbook.
  - Data Entry Basics
    - Simple and bulk entry methods along with the AutoFill feature.
  - Formatting Cells
    - Importance of formatting with numeric and text styles for clarity in data presentation.
  - Functions and Formulas
    - Use of basic arithmetic and built-in functions for calculations.
  - Data Sorting and Filtering
    - Techniques for managing large datasets effectively.
  - Data Analysis Tools
    - Features like Pivot Tables and What-If Analysis for deeper insights.
  - Protecting Your Spreadsheet
    - Techniques for safeguarding data through cell locking and password protection.
  - Cloud-Based Sharing and Collaboration
    - Real-time collaboration options in modern Excel versions.
  - Importing and Exporting Data
    - Features for importing data from various formats and exporting for other applications.
- Familiarisation with the Excel Workspace
  - Ribbon
    - Main control center organizing tools into tabs for easy access.
    - Tabs: Home, Insert, Data, View with specific functionalities.
  - Menus
    - Dropdown lists offering quick access to commands for specific tasks.
  - Toolbars
    - Quick Access Toolbar and Mini Toolbar for frequently used commands and formatting.
- Introduction to Data Types
  - Importance of Data Types
    - Efficient storage, data integrity, and usability in data manipulation.
  - Core Data Types in Excel
    - Text, Number, Date and Time, Boolean, and Error Types.
  - Advanced Data Types in Excel 365
    - Introduction of new data types powered by knowledge graph, like Stocks and Geography.
- Formatting Data in Excel
  - Importance of Formatting
    - Enhances readability, promotes professionalism, and improves efficiency.
  - Basic and Advanced Formatting Options
    - Various basic options include font styling, cell fill, borders, and number formatting.
    - Advanced options like conditional formatting and custom themes.
- Data Import and File Management
  - Techniques for Importing Data
    - Methods to import from CSV files, databases, web sources, and other Excel workbooks.
  - File Organization and Best Practices
    - Importance of naming conventions, folder structure, backup, and cleaning data.
- Summary
  - Recap of key learnings about Excel's functionality for data organization, analysis, visualization, and best practices for file management.